Class Overview

In Class:
- Midterm Discussion
- Dataflow Testing
- Half-Semester Project Brainstorming
- Midterm Exam Review

After Class:
- Office Hours: Symbolic Execution, Code Coverage Inputs, & Exam Questions

Homework:
- 2/18: Code Coverage Inputs
- 2/25: Midterm Assessment! (No Extensions or Late Submissions)
- 3/04: Half-Semester Project Idea

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Midterm Assessment Choices

You may *individually* select from:

- A "traditional" exam, or
- A presentation* on an allowed topic.

*No pre-recorded videos for the midterm assessment.

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Midterm Presentation Rubric

Content	**Worst**: Presentation does not focus on applicable content and has multiple factual errors. **Best**: Presentation is clearly focused on the topic, has examples, and has no factual errors.
Clarity	**Worst**: Not presented in an understandable or logical order and missing transitions between sections. **Best**: Concise presentation in a logical order with transitions between sections.
Sourcing	**Worst**: Based on only personal experience or assumptions, no citations included. **Best**: May interweave personal experience but is focused primarily on documented material with citations.
Materials	**Worst**: No presentation materials created. Presentation is ad-libbed or read directly from paper with no visual aids for the audience. **Best**: Effective visual aids directly related to content and examples covered.
Time	**Worst**: Significantly under (less than 6 minutes) or over (past 10 minutes) time. **Best**: Within the range of 7-9 minutes.

You will be required to submit any digital materials created for the presentation.

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Midterm Presentation Topics

- A/B Testing
- All-Pairs Testing
- API Testing
- Attack Patterns
- Burke-Fisher Error Repair
- Cause-Effect Graphs
- Characterization Test
- Classification Tree Method for Testing
- Concurrent Testing
- Differential Testing
- Elementary Comparison Testing
- Fagan Inspection
- Fuzzing
- Gray-Box Testing
- Happy Path Testing
- Heisenbug
- Probe Effect
- Reverse Semantic Traceability
- Runtime Error Detection
- Runtime Predictive Analysis
- Shift-Left Testing
- Testing Maturity Model
- Tree Testing

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Questions?

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CIS 613:
Decision Table Based Testing

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What are Decision Tables?

A declarative combination of:

- **Conditionals** – Boolean propositions (e.g., true/false, yes/no, 0/1) about the **current state** using environmental and system variables.
- **Rules** – Combinations of conditionals that correspond to **scenarios** with given solutions.
- **Actions** – Operations performed by the decision table below the level of abstraction of the decision table itself

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What are Decision Tables?

Conditionals

Rules

	Rule 1	Rule 2	Rule 3
Cond 1	T	F	T
Cond 2	T	T	T
Cond 3	T	F	F
Action 1	X		X
Action 2		X	
Action 3		X	
Action 4		X	
Action 5			X
Action 6			X
Action 7			X

Actions

General Properties:

- Represent Conditional Behavior
- Support *easy* analysis:
  - Consistency
  - Completeness
- Executable
- Declarative

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Decision Tables

How many rules would a decision table with 3 conditions have?

A. 4
B. 6
C. 8
D. 10

$8 = 2^n$, where n = 3

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Example

	Rule 1	*Rule 2*	*Rule 3*	*Rule 4*	*Rule 5*	*Rule 6*	*Rule 7*	*Rule 8*
c_1	T	T	T	T	F	F	F	F
c_2	T	T	F	F	T	T	F	F
c_3	T	F	T	F	T	F	T	F
a_1	X	X			X			
a_2	X					X		
a_3		X						
a_4			X	X	X		X	X

Condition c_3 has no effect on the actions of Rules 3 and 4. Similarly for Rules 7 and 8. They can be algebraically combined.

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Example

	Rule 1	*Rule 2*	***Rule 3,4***	*Rule 5*	*Rule 6*	***Rule 7,8***
c_1	T	T	T	F	F	F
c_2	T	T	F	T	T	F
c_3	T	F	-	T	F	-
a_1	X	X		X		
a_2	X				X	
a_3		X				
a_4			X	X		X

Condition c_3 has no effect on the actions of Rules 3 and 4. Similarly for Rules 7 and 8. They can be algebraically combined.

The '-' simply means 'don't care' or 'not applicable'

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Example

	Rule 1	*Rule 2*	***Rule 3,4,7,8***	*Rule 5*	*Rule 6*
c_1	T	T	-	F	F
c_2	T	T	F	T	T
c_3	T	F	-	T	F
a_1	X	X		X	
a_2	X				X
a_3		X			
a_4			X	X	

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Example: Rule Counting

	Rule 1	*Rule 2*	***Rule 3,4,7,8***	*Rule 5*	*Rule 6*
c_1	T	T	-	F	F
c_2	T	T	F	T	T
c_3	T	F	-	T	F
a_1	X	X		X	
a_2	X				X
a_3		X			
a_4			X	X	
Count					

- A rule (even with no don't care entries) counts as 1
- Each 'don't care' entry in a rule doubles the rule count
- For a table with n limited entry conditions, the sum of the rule counts should be 2^n.

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Example: Rule Counting

	Rule 1	*Rule 2*	***Rule 3,4,7,8***	*Rule 5*	*Rule 6*
c_1	T	T	-	F	F
c_2	T	T	F	T	T
c_3	T	F	-	T	F
a_1	X	X		X	
a_2	X				X
a_3		X			
a_4			X	X	
Count	1	1		1	1

- **A rule (even with no don't care entries) counts as 1**
- Each 'don't care' entry in a rule doubles the rule count
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Example: Rule Counting

	Rule 1	*Rule 2*	***Rule 3,4,7,8***	*Rule 5*	*Rule 6*
c_1	T	T	-	F	F
c_2	T	T	F	T	T
c_3	T	F	-	T	F
a_1	X	X		X	
a_2	X				X
a_3		X			
a_4			X	X	
Count	1	1	1 * 2 * 2 = 4	1	1

- A rule (even with no don't care entries) counts as 1
- **Each 'don't care' entry in a rule doubles the rule count**
- For a table with n limited entry conditions, the sum of the rule counts should be 2^n.

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Example: Rule Counting

	Rule 1	*Rule 2*	***Rule 3,4,7,8***	*Rule 5*	*Rule 6*
c_1	T	T	-	F	F
c_2	T	T	F	T	T
c_3	T	F	-	T	F
a_1	X	X		X	
a_2	X				X
a_3		X			
a_4			X	X	
Count	1	1	1 * 2 * 2 = 4	1	1

- A rule (even with no don't care entries) counts as 1
- Each 'don't care' entry in a rule doubles the rule count

- **For a table with n limited entry conditions, the sum of the rule counts should be 2^n.**

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Example: Last Day of the Month

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
c1. month in M1?	T	T	T	T	T	T	T	T	F	F	F	F	F	F	F	F
c2. month in M2?	T	T	T	T	F	F	F	F	T	T	T	T	F	F	F	F
c3. month in M3?	T	T	F	F	T	T	F	F	T	T	F	F	T	T	F	F
c4. leap year?	T	F	T	F	T	F	T	F	T	F	T	F	T	F	T	F
a1. last day = 30							X	X								
a2. last day = 31											X	X				
a3. last day = 28														X		
a4. last day = 29													X			
a5. impossible	X	X	X	X	X	X			X	X					X	X

M1 = { month : month has 30 days}
M2 = { month : month has 31 days}
M3 = { month : month is February}

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Example: Last Day of the Month

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
c1. month in M1?	T	T	T	T	T	T	T	T	F	F	F	F	F	F	F	F
c2. month in M2?	T	T	T	T	F	F	F	F	T	T	T	T	F	F	F	F
c3. month in M3?	T	T	F	F	T	T	F	F	T	T	F	F	T	T	F	F
c4. leap year?	T	F	T	F	T	F	T	F	T	F	T	F	T	F	T	F
a1. last day = 30							X	X								
a2. last day = 31											X	X				
a3. last day = 28														X		
a4. last day = 29													X			
a5. impossible	X	X	X	X	X	X			X	X					X	X

Rule pairs 1 and 2, 3 and 4, 5 and 6, 9 and 10 don't need c4, so they could be combined.

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Example: Last Day of the Month

	1	*2*	*3*	*4*	*5*	*6*	*7*	*8*	*9*	*10*	*11*	*12*	*13*	*14*	*15*	*16*
c1. month in M1?	T	T	T	T	T	T	T	T	F	F	F	F	F	F	F	F
c2. month in M2?	T	T	T	T	F	F	F	F	T	T	T	T	F	F	F	F
c3. month in M3?	T	T	F	F	T	T	F	F	T	T	F	F	T	T	F	F
c4. leap year?	T	F	T	F	T	F	T	F	T	F	T	F	T	F	T	F
a1. last day = 30							X	X								
a2. last day = 31											X	X				
a3. last day = 28														X		
a4. last day = 29													X			
a5. impossible	X	X	X	X	X	X			X	X					X	X

But this is impossible because c1, c2, and c3 are mutually exclusive.

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Example: Last Day of the Month (Extended Entry)

c1. month in	M1	M1	—	—	—	—
c2. month in	—	—	M2	M2	—	—
c3. month in	—	—	—	—	M3	M3
c4. leap year?	T	F	T	F	T	F
a1. last day = 30	x	x				
a2. last day = 31			x	x		
a3. last day = 28						x
a4. last day = 29					x	

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Example: Last Day of the Month (Extended Entry)

c1. month in	M1	M1	—	—	—	—
c2. month in	—	—	M2	M2	—	—
c3. month in	—	—	—	—	M3	M3
c4. leap year?	T	F	T	F	T	F
a1. last day = 30	x	x				
a2. last day = 31			x	x		
a3. last day = 28						x
a4. last day = 29					x	

But we can go further!

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Example: Last Day of the Month (Extended Entry)

c1. month in	M1	—	—	—
c2. month in	—	M2	—	—
c3. month in	—	—	M3	M3
c4. leap year?	—	—	T	F
a1. last day = 30	x			
a2. last day = 31		x		
a3. last day = 28				x
a4. last day = 29			x	

But we can go further!

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Example: Last Day of the Month (Extended Entry)

c1,2,3. month in	M1	M2	M3	M3
c4. leap year?	—	—	T	F
a1. last day = 30	x			
a2. last day = 31		x		
a3. last day = 28				x
a4. last day = 29			x	

M1 = { month : month has 30 days}
M2 = { month : month has 31 days}
M3 = { month : month is February}

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Example: Last Day of the Month (Extended Entry) Test Cases

	R_1	R_2	R_3	R_4
c1,2,3. month in	M1	M2	M3	M3
c4. leap year?	—	—	T	F
a1. last day = 30	x			
a2. last day = 31		x		
a3. last day = 28				x
a4. last day = 29			x	

	Month	Year	Last Day
Test Case for R_1	M1	Regular Year	30
Test Case for R_2	M2	Regular Year	31
Test Case for R_3	M3	Leap Year	29
Test Case for R_4	M3	Regular Year	28

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Example: Last Day of the Month (Extended Entry) Test Cases

	R_1	R_2	R_3	R_4
c1,2,3. month in	M1	M2	M3	M3
c4. leap year?	—	—	T	F
a1. last day = 30	x			
a2. last day = 31		x		
a3. last day = 28				x
a4. last day = 29			x	

	Month	Year	Last Day
Test Case for R_1	April	2019	30
Test Case for R_2	January	2019	31
Test Case for R_3	February	2020	29
Test Case for R_4	February	2019	28

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Impossible to Test

	1	*2*	*3*	*4*	*5*	*6*	*7*	*8*	*9*	*10*	*11*	*12*	*13*	*14*	*15*	*16*
c1. month in M1?	T	T	T	T	T	T	T	T	F	F	F	F	F	F	F	F
c2. month in M2?	T	T	T	T	F	F	F	F	T	T	T	T	F	F	F	F
c3. month in M3?	T	T	F	F	T	T	F	F	T	T	F	F	T	T	F	F
c4. leap year?	T	F	T	F	T	F	T	F	T	F	T	F	T	F	T	F
a1. last day = 30							X	X								
a2. last day = 31											X	X				
a3. last day = 28														X		
a4. last day = 29													X			
a5. impossible	X	X	X	X	X	X			X	X					X	X

M1 = { month : month has 30 days}
M2 = { month : month has 31 days}
M3 = { month : month is February}

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Combined Tests

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
c1. month in M1?	T	T	T	T	T	T	T	T	F	F	F	F	F	F	F	F
c2. month in M2?	T	T	T	T	F	F	F	F	T	T	T	T	F	F	F	F
c3. month in M3?	T	T	F	F	T	T	F	F	T	T	F	F	T	T	F	F
c4. leap year?	T	F	T	F	T	F	T	F	T	F	T	F	T	F	T	F
a1. last day = 30							X	X								
a2. last day = 31											X	X				
a3. last day = 28														X		
a4. last day = 29													X			
a5. impossible	X	X	X	X	X	X			X	X					X	X

M1 = { month : month has 30 days}
M2 = { month : month has 31 days}
M3 = { month : month is February}

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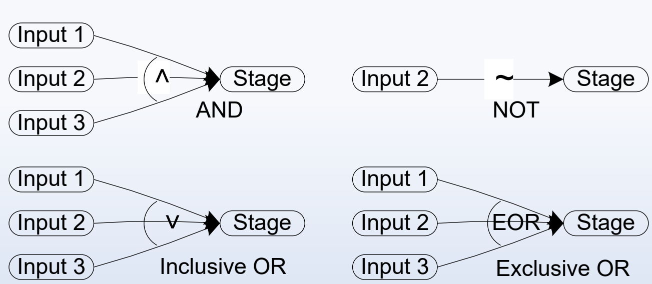
Where did Decision Tables Come From? Cause and Effect Graphs

Originally known as Cause and Effect Graphing

- Done with a graphical technique that expressed AND-OR-NOT logic.
- Causes and Effects were graphed like circuit components
- Inputs (conditions) to a circuit "caused" outputs (effects)



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